

Normalized Cartan closure, compensated perspective innovation, and triangular covariance frontier

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-091 left three named sufficient obligations. The projected Cartan FAR estimate was denoted H_C , the complete signed linear-rational NEAR estimate was denoted H_N , and the overlap/revisit-uniform progressive assembly was denoted H_A . This note advances all three without identifying them with the logically stronger full-OVERLAP theorem.

The main positive result is a closure of the *regular* Cartan branch, after correcting a transpose error in R-090 and R-091. The production Jacobian is not symmetric in general. Consequently the actual first-order coefficient b is not the spatial gradient of the endpoint coefficient c . This note replaces the false conservative compression by the exact R-089 two-tail trace and controls both b and $g = \nabla c$ with the same normalized whole-product estimate. No conclusion below uses $b = \nabla c$.

For the R-079 regular mutually orthogonal, strict-past, no-revisit, one-shot whole-shell control class and every $C \geq 5$,

$$\boxed{\mathcal{H}_C^{\text{reg}}(C) \leq C_{e,L} 2^{-(C-5)/2} \{C_{\varepsilon,\delta,e,L} + \varepsilon \mathbb{E}X + \delta \mathbb{E}Y\}.} \quad (1.1)$$

Here $e = \epsilon_\rho > 0$ is fixed, X is the accepted Cameron–Martin energy, and $Y = \|U + A^*\|_6^6$. The constants are cutoff-uniform. The proof uses a normalized lift of the production quotient. Its Jacobian is a finite polynomial of degree at most four in a bounded vector, so the infinite rational Taylor forest disappears before the FAR estimate. A fractional largest-input argument gives the positive output exponent, and the physical sharp-shell square function controls every regular prefix exactly once.

The second result compresses every R-091 conditional companion in H_N into one matrix-perspective martingale telescope. After the already required baseline endpoint is restored, the only intrinsically signed high-root density is

$$\tilde{\mathcal{C}}_k = \mathcal{K}_k + P_{k-1}|d_k y|^2 - B_{k-1} : \Delta \Gamma_k. \quad (1.2)$$

The terminal $\Theta_R(B_1)$ energy supplies the new Doob term and cannot be discarded. A coefficient-conditioned moment-matched one-reveal branch is nonnegative; in general the sign compresses to a weighted conditional- covariance deficit plus coefficient-reveal variances. An exact finite fixture makes the fully completed local expression equal $-623/5440$, so no fixed or newly chosen adaptive derivative payment creates automatic termwise positivity.

The third result proves the exact covariance-union contraction and a causal entropy-union identity for arbitrary overlapping temporal source ranges. Overlap itself creates no multiplicity constant, and kernel/revisit cost is retained as a positive fibre-entropy surplus. The polar-minimal source representative is generally nonadapted, while the entropy identity avoids that representative. Its free-energy term is exactly the still-open Nelson quantity, so the H_A successor remains a triangular covariance-union signed-packet theorem in source coordinates.

Finally, two tempting global shortcuts are audited. A negative flow removes the one-shot Ramer determinant zero but replaces it by a high-field non-surjectivity complement and a signed material derivative.

Conditional CAT(0) barycentering alone fails on a scaled reset model for every fixed allocation. These are method no-gos, not production counterexamples.

The result does not prove the complete signed H_N estimate, the triangular H_A estimate, general progressive/revisit Cartan control, REG as a complete packet theorem, full OVERLAP, the $q = 10/9$ Nelson estimate, an interacting measure, or Sector-A closure. Tier stays T4.

The load-bearing authorities are R-075, R-079, R-084, R-087, and R-089–R-091. R-075 supplies the physical sharp-shell graph norm. R-079 supplies the exact regular control class and complete current. R-084 supplies the root OU identity. The exact two-field trace is the R-089 trace retained in R-090's predecessor data. R-092 audits R-090's conservative identity and the corresponding second equality in R-091 as false outside a symmetric Jacobian slice. R-091's first, definition-level nonnegative output block, lossless weight extraction, and terminal nonduplication identities remain valid.

2 The normalized production Jacobian is a finite forest

Let $S = S^T$, $\|S\| \leq 1$, and put

$$d_x = e + |x|^2, \quad n_x = \frac{x}{\sqrt{d_x}}, \quad q_x = n_x^T S n_x, \quad P_x = n_x n_x^T. \quad (2.1)$$

The production quotient vector is

$$F_S(x) = q_x x. \quad (2.2)$$

Theorem 2.1 (finite normalized Jacobian)

In the fixed six-real convention,

$$J_S(x) := DF_S(x) = q_x I + 2P_x(S - q_x I) = q_x I + 2P_x S - 2q_x P_x. \quad (2.3)$$

Consequently J_S is a polynomial of degrees two and four in the bounded variable n_x , and

$$J_S(x)^T - J_S(x) = 2(SP_x - P_x S). \quad (2.4)$$

In particular, the Cartan antisymmetric part is the quadratic commutator $SP_x - P_x S$.

Proof. Since

$$Dq_x[h] = \frac{2}{\sqrt{d_x}}(S - q_x I)n_x \cdot h,$$

one has $DF_S(x)[h] = q_x h + 2n_x n_x^T(S - q_x I)h$, which is (2.3). Taking the transpose cancels both scalar terms and gives (2.4). The accepted production variance identity also gives $\|(S - q_x I)n_x\| \leq \|n_x\|$, hence $\|J_S(x)\| \leq 3$. $\square$

For $x_- = x$, $x_+ = x + a$, write

$$n_{\pm} = n_e(x_{\pm}), \quad \delta n = n_+ - n_-, \quad \Delta q = q_+ - q_-, \quad \Delta P = P_+ - P_-. \quad (2.5)$$

Then

$$\Delta q = (n_+ + n_-)^T S \delta n, \quad \Delta P = \delta n n_+^T + n_- \delta n^T, \quad (2.6)$$

and

$$\Delta J = \Delta q I + 2\Delta P S - 2\Delta q P_+ - 2q_- \Delta P. \quad (2.7)$$

Every term in (2.7) contains exactly one δn and at most three other bounded normalized factors. Thus

$$\|\Delta J\| \leq 14|\delta n|. \quad (2.8)$$

The antisymmetric increment is even simpler:

$$\frac{\Delta(J^T - J)}{2} = S\Delta P - \Delta P S. \quad (2.9)$$

3 Exact normalized secant and saturation

The normalized difference has the exact resolvent formula

$$\delta n = \left[\frac{I}{\sqrt{d_+}} - \frac{x(x_+ + x_-)^T}{\sqrt{d_- d_+}(\sqrt{d_-} + \sqrt{d_+})} \right] a. \quad (3.1)$$

Indeed, subtract $x/\sqrt{d_-}$ from $(x + a)/\sqrt{d_+}$ and use

$$\sqrt{d_+} - \sqrt{d_-} = \frac{a \cdot (x_+ + x_-)}{\sqrt{d_+} + \sqrt{d_-}}.$$

There is also a coordinate-free chord proof of the sharp useful bound. Lift x to $X = (x, \sqrt{e})$ and y to $Y = (y, \sqrt{e})$. For nonzero vectors,

$$\left| \frac{X}{|X|} - \frac{Y}{|Y|} \right| \leq \frac{2|X - Y|}{|X| + |Y|}. \quad (3.2)$$

After projecting onto the first six coordinates and taking $y = x + a$,

$$|\delta n| \leq 2 \min \left\{ 1, \frac{|a|}{\sqrt{d_-} + \sqrt{d_+}} \right\} \leq \min \{ 2, e^{-1/2} |a| \}. \quad (3.3)$$

This is the vector and multimode saturation missing from the cumulative Z^6 majorant rejected in R-091.

For a centered target-value heat variable $\zeta \sim N(0, \Sigma)$, couple the same ζ at both endpoints. Then

$$D(P_\Sigma F_S)(x) = \mathbb{E}_\zeta J_S(x + \zeta). \quad (3.4)$$

If

$$c_\Sigma(x, a) = P_\Sigma F_S(x + a) - P_\Sigma F_S(x),$$

then $|c_\Sigma| \leq 3|a|$ and the exact chain rule is

$$\partial_i c_\Sigma = \mathbb{E}_\zeta [\Delta J(x + \zeta, x + a + \zeta) \partial_i x + J_S(x + a + \zeta) \partial_i a]. \quad (3.5)$$

The actual R-084/R-089 current coefficient instead is

$$b_i = \mathbb{E}_\zeta [\Delta J(x + \zeta, x + a + \zeta)^T \partial_i x + J_S(x + a + \zeta)^T \partial_i a]. \quad (3.6)$$

Therefore, with $g = \nabla c_\Sigma$,

$$|b| + |g| \leq 6|Da| + 28\mathbb{E}_\zeta[|\delta n| |Dx|]. \quad (3.7)$$

No complete translated $|x|^6$ multiplier occurs.

The transpose cannot be removed. In fact

$$b_i - g_i = \mathbb{E}_\zeta [(\Delta J^T - \Delta J) \partial_i x + (J_+^T - J_+) \partial_i a], \quad (3.8)$$

and (2.5) shows that the right side is a generally nonzero Cartan commutator. This is the audit correction to R-090 (2.4) and R-091 (2.1).

For later fractional estimates it is important not to insert a spurious $|a||Da|$ term. The exact identity is

$$D\delta n = Dn_e(x + a)Da + \{Dn_e(x + a) - Dn_e(x)\}Dx. \quad (3.9)$$

Since $\|Dn_e\| \leq e^{-1/2}$ and $\|D^2 n_e\| \leq C_e$,

$$|D\delta n| \leq C_e\{|Da| + |a||Dx|\}. \quad (3.10)$$

4 Exact two-field output trace and the transpose audit

The equality claimed in R-090 (2.4) is false for a nonsymmetric production Jacobian. Retain instead the exact R-089 first-order form

$$(L_{A,j,t}v)_i = b_{A,j,t,i}^T v + c_{A,j,t}^T \partial_i v.$$

For a root mode $v_{p,\mu} = \sqrt{\lambda_{j,p}} e^{ipx} e_\mu$, put $r = q - p$. The definition-level nonnegative output block is exactly

$$\begin{aligned} \mathcal{E}_{A,j,m} &:= \int_0^\infty e^{-2t} \mathbb{E} \sum_{\mu,p} \lambda_{j,p} \sum_q |\pi_m(q)|^2 \\ &\quad \times \sum_i \left| \widehat{b}_{A,j,t,i}(r)^T e_\mu + ip_i \widehat{c}_{A,j,t}(r)^T e_\mu \right|^2 dt. \end{aligned} \quad (4.1)$$

This is the first equality in the R-091 block definition. Its replacement there by a single q -weighted c trace is the audited false step.

Let $g_{A,j,t} = \nabla c_{A,j,t}$. Suppose $|p|_\infty \leq 2^j$, $|q|_\infty > 2^{m-1}$, and $m \geq j + 5$. Then

$$|r|_\infty > 15 \cdot 2^j. \quad (4.2)$$

The enlarged coefficient projector used in R-089 is supported where $|r|_\infty > 2^{m-2}$. Consequently Plancherel gives the safe coefficient-band Poincare estimate

$$\|\widetilde{\Pi}_m c\|_2^2 \leq 16 \cdot 2^{-2m} \|\widetilde{\Pi}_m g\|_2^2. \quad (4.3)$$

Using $|u + v|^2 \leq 2|u|^2 + 2|v|^2$ and the pinned root moments

$$\Lambda_{0,j} \leq \kappa_0 2^{-j}, \quad \Lambda_{1,j} \leq \kappa_1 2^j,$$

the exact trace obeys

$$\boxed{\begin{aligned} \mathcal{E}_{A,j,m} &\leq \int_0^\infty e^{-2t} \mathbb{E} \left[2\kappa_0 2^{-j} \|\widetilde{\Pi}_m b_{A,j,t}\|_2^2 \right. \\ &\quad \left. + 32\kappa_1 2^{-j} 2^{-2(m-j)} \|\widetilde{\Pi}_m g_{A,j,t}\|_2^2 \right] dt. \end{aligned}} \quad (4.4)$$

Thus the omitted antisymmetric remainder is not discarded: it stays in b . The derivative channel is paid separately through g , and its current-root covariance carries the additional gap factor $2^{-2(m-j)}$.

5 Fractional normalized-lift and whole-product estimate

Lemma 5.1 (bounded fractional lift)

Let $0 < \sigma < \theta < 1$, $p\theta > 1$, and let u be spatially differentiable on the fixed torus. Then

$$\boxed{[n_e(u)]_{W^{\sigma,p}} \leq C_{e,L,\sigma,\theta,p} \|Du\|_{L^{p\theta}}^\theta.} \quad (5.1)$$

If $\delta n = n_e(x+a) - n_e(x)$, then

$$\boxed{[\delta n]_{W^{\sigma,p}} \leq C_{e,L,\sigma,\theta,p} \left\{ \|Da\|_{L^{p\theta}}^\theta + \|a\| \|Dx\|_{L^{p\theta}}^\theta \right\}.} \quad (5.2)$$

Proof. The chord bound gives

$$|n_e(u(x+h)) - n_e(u(x))| \leq C \min\{1, e^{-1/2} |\delta_h u(x)|\} \leq C_e |\delta_h u(x)|^\theta. \quad (5.3)$$

The translation estimate $\|\delta_h u\|_{p\theta} \leq |h| \|Du\|_{p\theta}$ and the fractional difference quotient converge exactly because $\theta > \sigma$. This proves (5.1). Apply the same bounded-function argument to δn and use (3.9)–(3.10) to prove (5.2). $\square$

Let Dx , Da have physical Fourier support in the sharp cube Q_j , and let $m \geq j + 5$. For a fixed target-value heat sample ζ , which has no spatial derivative, set

$$N_{\pm} = [n_e(x_{\pm} + \zeta)]_{W^{\sigma,p}}, \quad D_{\sigma} = [\delta n]_{W^{\sigma,p}}, \quad D_{\infty} = \|\delta n\|_{\infty}, \quad (5.4)$$

and choose q by $1/2 = 1/p + 1/q$. Because (2.3) and (2.7) contain only finitely many bounded normalized factors, the fractional product rule gives

$$[J_+]_{W^{\sigma,p}} \leq CN_+, \quad [\Delta J]_{W^{\sigma,p}} \leq C\{D_{\sigma} + D_{\infty}(N_+ + N_-)\}. \quad (5.5)$$

All Fourier convolutions are retained before this bound is applied. A high output of a product with a Q_j factor must therefore come from the coefficient, and

$$\|\tilde{\Pi}_m b_{A,j,t}\|_2^2 + \|\tilde{\Pi}_m g_{A,j,t}\|_2^2 \leq C_e 2^{-2\sigma m} \mathbb{E}_{\zeta} \Re_{A,j,t}(\zeta)^2, \quad (5.6)$$

where

$$\begin{aligned} \Re_{A,j,t}(\zeta) = & N_+ \|Da\|_q \\ & + \{D_{\sigma} + D_{\infty}(N_+ + N_-)\} \|Dx\|_q. \end{aligned} \quad (5.7)$$

Before squaring, each heat-averaged field is bounded by $C_e 2^{-\sigma m} \mathbb{E}_{\zeta} \Re(\zeta)$; Jensen gives (5.6). The same proof applies to b , which uses transposed coefficient matrices, and g , which uses untransposed matrices; transposition preserves every operator and fractional coefficient norm. This is a whole-product largest-input estimate. It assumes neither output orthogonality nor scalar alignment and includes coherent high-high-to-low collisions.

6 Regular Cartan H_C closes

Let $\mathcal{B}_{\gamma}^{\text{out}}$ be the exact R-091 weighted output ledger. Use its definition-level block (4.1), insert (5.6) into the corrected two-field estimate (4.4), multiply output offset $m - j - 5$ by $2^{2\gamma(m-j-5)}$, and sum first in m . For $0 < \gamma < \sigma$,

$$\mathcal{B}_{\gamma}^{\text{out}} \leq C_e \left\{ \frac{2\kappa_0 2^{-10\sigma}}{1 - 2^{-2(\sigma-\gamma)}} + \frac{32\kappa_1 2^{-10(1+\sigma)}}{1 - 2^{-2(1+\sigma-\gamma)}} \right\} \sum_{A,j} 2^{-(1+2\sigma)j} \mathbb{E} \int_0^{\infty} e^{-2t} \mathbb{E}_{\zeta} \Re_{A,j,t}(\zeta)^2 dt. \quad (6.1)$$

Choose

$$\gamma = \frac{1}{4}, \quad \sigma = \frac{4}{15}, \quad \theta = \frac{3}{10}, \quad p = 6, \quad q = 3. \quad (6.2)$$

Then $p\theta = 9/5 > 1$, $\sigma < \theta$, and

$$\begin{aligned} 2^{-(1+2\sigma)j} &= 2^{-23j/15}, \\ 1 - 2^{-2(\sigma-\gamma)} &= 1 - 2^{-1/30} > 0, \\ 1 - 2^{-2(1+\sigma-\gamma)} &= 1 - 2^{-61/30} > 0. \end{aligned} \quad (6.3)$$

For the cutoff Gaussian prefix, every fixed moment of $\|DU_j\|_{L^r}$ is bounded by a constant times $2^{j/2}$. Expanding (5.7) with (5.1)–(5.2), the largest Gaussian power is $2 + 2\theta = 13/5$, hence the worst shell growth is

$$2^{(13/5)j/2} = 2^{13j/10}. \quad (6.4)$$

The remaining geometric factor is strictly summable:

$$\boxed{2^{-23j/15} 2^{13j/10} = 2^{-7j/30}.} \quad (6.5)$$

Write for a control prefix

$$X_a = \|a\|_{H^2}^2, \quad Y_a = \|a\|_6^6.$$

The interpolation $\|Da\|_3^2 \lesssim X_a^{1/2} Y_a^{1/6}$ and the normalized-lift powers in (5.1)–(5.2) give an exhaustive ledger after $Dx = DU + Da$ and $2uv \leq u^2 + v^2$ are used. Constants and Gaussian factors are suppressed in the middle column:

source in $\mathfrak{R}^2$	control monomial	total exponent
$\ Da\ _3^2$	$X_a^{1/2} Y_a^{1/6}$	2/3
$N_{+, \text{ctrl}}^2 \ Da\ _3^2$	$X_a^{4/5} Y_a^{1/6}$	29/30
fractional Da pieces	$X_a^{3/10}$	3/10
$\ aDa\ _{9/5}^{2\theta} \ Da\ _3^2$	$X_a^{3/5} Y_a^{1/3}$	14/15
$\ aDU\ _{9/5}^{2\theta} \ Da\ _3^2$	$X_a^{1/2} Y_a^{4/15}$	23/30
$\ aDa\ _{9/5}^{2\theta} \ DU\ _3^2$	$X_a^{1/10} Y_a^{1/6}$	4/15
$\ aDU\ _{9/5}^{2\theta} \ DU\ _3^2$	$Y_a^{1/10}$	1/10

(6.6)

Indeed $N_{+, \text{ctrl}}^2 \lesssim X_a^{3/10}$. For the terms in (5.2), put $r = p\theta = 9/5$, choose s by $1/r = 1/6 + 1/s$, and set

$$s = \frac{6\theta}{1 - \theta} = \frac{18}{7}, \quad \alpha = 2 - \frac{1}{2\theta} = \frac{1}{3}.$$

The three-dimensional Gagliardo–Nirenberg and Holder inequalities give

$$\|Da\|_s \lesssim \|a\|_{H^2}^\alpha \|a\|_6^{1-\alpha}, \quad \|aDa\|_r^{2\theta} \lesssim X_a^{\theta\alpha} Y_a^{\theta(2-\alpha)/3} = X_a^{1/10} Y_a^{1/6}.$$

For the inner aDU branch, the cutoff Gaussian factor has every fixed moment and the control factor is only $\|a\|_6^{2\theta} = Y_a^{\theta/3} = Y_a^{1/10}$. Multiplying each inner branch by the outer DU or Da , and using $2uv \leq u^2 + v^2$, gives all four rows displayed above. Thus every cross term is dominated by two entries of the table. The largest total power is

$$\frac{4}{5} + \frac{1}{6} = \frac{29}{30} < 1. \quad (6.7)$$

All mixed Gaussian factors are handled pointwise by generalized Young and then by the accepted Gaussian moments. Thus every allocation $\varepsilon, \delta > 0$ is admissible.

It remains to justify that a prefix uses the terminal budgets only once. This is true on the exact regular class and is false as a formal inference for arbitrary revisits. Write

$$A^* = A^0 + \sum_{k=j_0}^J a_k, \quad a_k = K_k h_k, \quad A^{(j)} - A^0 = \sum_{k=j_0}^j a_k. \quad (6.8)$$

The last sum is the *physical sharp-shell* prefix $P_{[j_0, j]}^{\text{phys}}(A^* - A^0)$; it is not identified with the auxiliary smooth analytic LP used to test outputs. R-075 (8.1)–(8.4) gives

$$\boxed{\begin{aligned} \|A^{(j)} - A^0\|_6 &\lesssim \|S_j h\|_6 \leq \|S_J h\|_6 \lesssim \|A^* - A^0\|_6, \\ \|A^{(j)} - A^0\|_{H^2}^2 &\lesssim \sum_{k \leq j} \|h_k\|_{\mathcal{H}_k}^2 \leq X. \end{aligned}} \quad (6.9)$$

Adaptation does not affect these pathwise spatial inequalities. The fixed low object is paid by R-080, while $\|A^*\|_6^6 \lesssim Y + \|U\|_6^6$ and the Gaussian term has a uniform moment. Therefore (6.1)–(6.9) prove

$$\boxed{\mathcal{B}_{1/4}^{\text{out}} \leq C_{e,L} \{C_{\varepsilon,\delta,e,L} + \varepsilon \mathbb{E}X + \delta \mathbb{E}Y\}.} \quad (6.10)$$

R-091's lossless output extraction now yields (1.1):

$$\mathcal{H}_C^{\text{reg}}(C) \leq 2^{-(C-5)/2} \mathcal{B}_{1/4}^{\text{out}}. \quad (6.11)$$

This closes the regular Cartan FAR obligation. It does not close a progressive revisit prefix. For example, the temporal source sequence $tf, -tf$ has zero terminal shift but a prefix of size $t\|f\|_6$. The terminal Douglas contraction controls its Cameron–Martin endpoint, not this intermediate L^6 norm. A different signed temporal argument is required inside H_A .

7 Vector tail boundary

R-091's scalar one-mode superexponential tail is not uniform over vector ellipticity. At zero floor, take

$$S = \text{diag}(1, -1), \quad v_\epsilon(x) = (\sin Nx, \epsilon \cos Nx).$$

The denominator of the homogeneous quotient has the Fourier-geometric ratio

$$\rho_\epsilon = \frac{1 - \epsilon}{1 + \epsilon} \longrightarrow 1 \quad (\epsilon \downarrow 0). \quad (7.1)$$

At positive floor the same boundary appears above the scale $A\epsilon \gtrsim \sqrt{e}$. Hence no scalar-uniform exponential ratio may be inserted into the vector proof. The whole-product fractional estimate (5.6), which is uniform in direction and allows coherent collisions, is the correct repair. This fires the narrowly scoped method result NG-2026-07-25-A13-SCALAR-SUPEREXPONENTIAL-VECTOR-UNIFORMITY.

8 Exact conditional perspective compression

Let P_k be a finite filtration and consider the conditional endpoint

$$\mathcal{E} = \frac{1}{2}D : Q + G^T C c + \frac{1}{2}c^T B c, \quad Q = GG^T - \Gamma, \quad (8.1)$$

with $c \in \mathcal{F}_\ell$. Allow one terminally allocated matrix payment $c^T R c$, $R \succ 0$, and set

$$x = C^T G, \quad A = B + 2R, \quad x_k = P_k x, \quad A_k = P_k A, \quad m_k = A_k^{-1} x_k. \quad (8.2)$$

Completing the conditional square gives

$$\boxed{P_\ell[\mathcal{E} + c^T R c] = \frac{1}{2}\|c + m_\ell\|_{A_\ell}^2 + \frac{1}{2}P_\ell[D : Q] - \frac{1}{2}x_\ell^T A_\ell^{-1} x_\ell.} \quad (8.3)$$

The R-091 same-root companions are the exact martingale cross-brackets

$$\boxed{r_C = P_\ell \sum_{k>\ell} (d_k C)^T (d_k G), \quad 2J_D = P_\ell \sum_{k>\ell} d_k D : d_k Q.} \quad (8.4)$$

Thus they are not independent error terms.

Theorem 8.1 (matrix-perspective telescope)

Define $\Phi_k = x_k^T A_k^{-1} x_k$. Then

$$P_{k-1}\Phi_k - \Phi_{k-1} = \mathcal{K}_k, \quad \mathcal{K}_k = P_{k-1}[(m_k - m_{k-1})^T A_k (m_k - m_{k-1})] \geq 0. \quad (8.5)$$

Consequently

$$2\mathfrak{S}_{\ell,R} = P_\ell \left[D : Q - x^T A^{-1} x + \sum_{k>\ell} \mathcal{K}_k \right]. \quad (8.6)$$

Proof. Expand the square in (8.5). Conditional expectation gives $P_{k-1}(A_k m_k) = x_{k-1}$ and $P_{k-1}A_k = A_{k-1}$. The cross and old-square terms therefore reduce to $x_{k-1}^T A_{k-1}^{-1} x_{k-1}$. Telescoping and (8.3) prove (8.6). $\square$

The positive $\mathcal{K}_k$ terms partition the terminal perspective endpoint. Appending them to that endpoint as new positivity would count the same square twice.

9 Baseline restoration and the augmented perspective density

For the complete linear-rational frame,

$$D = B_1 - B_0, \quad C = B = B_1. \quad (9.1)$$

Put

$$\Theta_R(B) = B - B(B + 2R)^{-1}B \succeq 0. \quad (9.2)$$

After restoring the once-only baseline $W_0 = \frac{1}{2}B_0 : Q$, (8.3)–(8.6) become

$$\begin{aligned} P_\ell[W_0 + \Delta W_{LR} + c^T R c] &= \frac{1}{2} \|c + m_\ell\|_{A_\ell}^2 \\ &\quad + \frac{1}{2} P_\ell \left[G^T \Theta_R(B_1) G - B_1 : \Gamma + \sum_{k>\ell} \mathcal{K}_k \right]. \end{aligned} \quad (9.3)$$

The apparent $-B_0$ Schur debt has disappeared because it was the baseline endpoint, not a new local error.

Let $\Gamma = \Gamma_\ell + \sum_{k>\ell} \Delta\Gamma_k$ and $B_k = P_k B_1$. Conditional chronology gives

$$P_\ell[B_1 : \Gamma] = B_\ell : \Gamma_\ell + P_\ell \sum_{k>\ell} B_{k-1} : \Delta\Gamma_k. \quad (9.4)$$

Therefore

$$\begin{aligned} P_\ell[W_0 + \Delta W_{LR} + c^T R c] &= \frac{1}{2} \|c + m_\ell\|_{A_\ell}^2 + \frac{1}{2} P_\ell[G^T \Theta_R(B_1) G] \\ &\quad - \frac{1}{2} B_\ell : \Gamma_\ell + \frac{1}{2} P_\ell \sum_{k>\ell} \mathcal{C}_k^{\text{Schur}}, \end{aligned} \quad (9.5)$$

where

$$\mathcal{C}_k^{\text{Schur}} = \mathcal{K}_k - B_{k-1} : \Delta\Gamma_k. \quad (9.6)$$

This intermediate density is not the smallest inseparable object. Define

$$Y = \Theta_R(B_1)^{1/2} G, \quad y_k = P_k Y.$$

Doob Pythagoras gives

$$P_\ell |Y|^2 = |y_\ell|^2 + P_\ell \sum_{k>\ell} P_{k-1} |d_k y|^2. \quad (9.7)$$

Hence (9.5) is exactly

$$\boxed{P_\ell [W_0 + \Delta W_{LR} + c^T R c] = \frac{1}{2} \|c + m_\ell\|_{A_\ell}^2 + \frac{1}{2} |y_\ell|^2 - \frac{1}{2} B_\ell : \Gamma_\ell + \frac{1}{2} P_\ell \sum_{k>\ell} \tilde{C}_k,} \quad (9.8)$$

where

$$\boxed{\tilde{C}_k = \mathcal{K}_k + P_{k-1} |d_k y|^2 - B_{k-1} : \Delta \Gamma_k.} \quad (9.9)$$

For frozen predictable B, R and a fresh covariance increment, $\mathcal{C}_k^{\text{Schur}} = -\Theta_R(B) : \Delta \Gamma_k \leq 0$, whereas $P_{k-1} |d_k y|^2 = \Theta_R(B) : \Delta \Gamma_k$ and $\tilde{C}_k = 0$. Thus the terminal positive energy is load-bearing and cannot merely be set aside.

There is an exact one-reveal criterion. Let $\mathcal{F}$ be the past, $G = G_0 + g$, let $B \succeq 0$ be revealed at the new level, and let $R \succ 0$ be $\mathcal{F}$ -measurable. Put

$$\begin{aligned} \mu_B &= \mathbb{E}[g \mid \mathcal{F}, B], & V_B &= \text{Cov}(g \mid \mathcal{F}, B), \\ A &= B + 2R, & \bar{B} &= \mathbb{E}[B \mid \mathcal{F}], & \bar{A} &= \bar{B} + 2R, \\ u_B &= A^{-1} B (G_0 + \mu_B), & m_0 &= \bar{A}^{-1} \mathbb{E}[B (G_0 + \mu_B) \mid \mathcal{F}], \\ h_B &= \Theta_R(B)^{1/2} (G_0 + \mu_B). \end{aligned} \quad (9.10)$$

For $m = A^{-1} B (G_0 + g)$ and $Y = \Theta_R(B)^{1/2} (G_0 + g)$, direct conditional expansion and $BA^{-1}B + \Theta_R(B) = B$ yield

$$\boxed{\begin{aligned} &\mathbb{E}[(m - m_0)^T A (m - m_0) + |Y - \mathbb{E}[Y \mid \mathcal{F}]|^2 \mid \mathcal{F}] - \bar{B} : \Delta \Gamma \\ &= \mathbb{E}[(u_B - m_0)^T A (u_B - m_0) \mid \mathcal{F}] + \text{Var}_{\mathcal{F}}(h_B) + \mathbb{E}[B : V_B \mid \mathcal{F}] - \bar{B} : \Delta \Gamma. \end{aligned}} \quad (9.11)$$

In particular, if $\mu_B = 0$ and $V_B = \Delta \Gamma$ independently of B , the augmented one-step density is nonnegative. This closes the frozen and coefficient-conditioned moment-matched value-only branch. It does not cover the multistep production derivative-feedback correlation.

The coefficient bracket r_C is encoded in $x_k = P_k(C^T G)$ and hence in the perspective telescope. The quadratic bracket J_D remains in $P_\ell[D : Q]$ until the full-frame baseline identity is used; it is not an extra copy of $\mathcal{K}_k$. Heat is contained in the chosen terminal coefficient martingale, and the R-063 forest reconstructs $B_1 : Q$ rather than adding another endpoint.

10 Perspective positivity fails, but the target sharpens

For $R = \eta I$, the minimized local remainder is

$$\mathfrak{S}_\eta = \kappa_\ell - \frac{1}{2} h^T (\hat{B} + 2\eta I)^{-1} h, \quad \kappa_\ell = \frac{1}{2} P_\ell[D : Q]. \quad (10.1)$$

It is increasing and concave in η :

$$\mathfrak{S}'_\eta = h^T (\hat{B} + 2\eta I)^{-2} h, \quad \mathfrak{S}''_\eta = -4h^T (\hat{B} + 2\eta I)^{-3} h. \quad (10.2)$$

If $\kappa_\ell < 0$, no derivative payment restores a local sign. Choosing fresh R_k at each filtration level is not a cure: the telescope acquires

$$x_{k-1}^T (\bar{A}_k^{-1} - A_{k-1}^{-1}) x_{k-1}, \quad \bar{A}_k = P_{k-1} A_k, \quad (10.3)$$

with uncontrolled sign unless $R_{k-1} \succeq P_{k-1}R_k$.

An exact finite fixture contains all conditional companions. Let $H \in \{0, 1\}$ equiprobably and set

$$V_0 = \frac{1}{2}, \quad V_1 = \frac{3}{2}, \quad \Gamma = 1.$$

Conditionally on H , let $G = 0$ with probability $1/2$ and $G = \pm\sqrt{2V_H}$ with probability $1/4$ each. Put

$$R_G = G^2 - V_H, \quad m_H = -\frac{1}{4}(V_H - 1), \quad (10.4)$$

$$B_0 = 1, \quad B_1 = C = B = 1 + m_H - \frac{1}{8}R_G + \frac{1}{8}G, \quad D = B_1 - B_0, \quad \eta = \frac{1}{2}. \quad (10.5)$$

Then $B_1 \geq (11 - 2\sqrt{3})/16 > 0$,

$$r_C = \frac{V_H}{8}, \quad J_D = -\frac{V_H^2}{16}, \quad (10.6)$$

and the exact conditional minima are

$$\mathfrak{S}_0 = -\frac{13}{272}, \quad \mathfrak{S}_1 = -\frac{29}{160}. \quad (10.7)$$

Consequently

$$\mathbb{E} \left[\inf_{c \in L^2(\mathcal{F}_H)} \{\mathcal{E} + \eta c^2\} \right] = -\frac{623}{5440} < 0. \quad (10.8)$$

The infimum is attained by the explicit branchwise optimizer from the completed square in (8.3). The covariance is matched after averaging, while covariance mismatch, r_C , J_D , and Schur debt are all nonzero conditionally. This is an algebraic no-go, not a post-paid production counterexample. It fires NG-2026-07-25-A13-PERSPECTIVE-INNOVATION-TERMWISE-POSITIVITY.

The moment-matching premise in (9.11) is essential. With trivial past, $G_0 = 0$, $B = 1, 2$ equiprobably, and centered conditional increments of variances $2, 1$ respectively, $\Delta\Gamma = 3/2$ but

$$\mathbb{E}[BV_B] - \mathbb{E}[B]\Delta\Gamma = -\frac{1}{4}. \quad (10.9)$$

Both positive coefficient-reveal terms in (9.11) vanish, for every $R \succ 0$. Thus a weighted conditional-covariance deficit is the genuine remaining sign mechanism.

The sharp nonduplicating H_N successor is a form bound for

$$\Lambda_{\text{low-paid}} + \frac{1}{2}|y_\ell|^2 - \frac{1}{2}B_\ell : \Gamma_\ell + \frac{1}{2} \sum_{k>\ell} \tilde{C}_k. \quad (10.10)$$

The completion square is retained. Formula (9.11) proves the moment-matched one-reveal subcase and compresses the general failure to a weighted conditional-covariance deficit plus coefficient-reveal terms. No current authority proves (10.10) for the full multistep derivative-feedback packet, so H_N remains open.

11 Exact covariance union and its causal boundary

Let $\mathcal{K} = \bigoplus_{k=1}^N \mathcal{K}_k$ and let $S_k : \mathcal{K}_k \rightarrow H$ be finite-dimensional source blocks. Define

$$S_\pi(h_1, \dots, h_N) = \sum_k S_k h_k, \quad C = S_\pi S_\pi^* = \sum_k S_k S_k^*. \quad (11.1)$$

Theorem 11.1 (covariance-union contraction)

For standard independent $\xi_k \in \mathcal{K}_k$, $X = \sum_k S_k \xi_k$ is centered Gaussian with covariance C , and

$$\|C^{\dagger/2} S_{\pi} h\|^2 = \|P_{(\ker S_{\pi})^{\perp}} h\|^2 \leq \sum_k \|h_k\|^2. \quad (11.2)$$

For every common terminal operator L ,

$$\sum_k \|L S_k\|_{\text{HS}}^2 = \text{Tr}(L C L^*) = \|L C^{1/2}\|_{\text{HS}}^2. \quad (11.3)$$

Proof. The covariance identity follows from independence. The polar decomposition $S_{\pi} = C^{1/2} R$ makes $C^{\dagger/2} S_{\pi}$ the orthogonal projection of R onto $\text{Ran } C$, proving (11.2). Taking the Hilbert–Schmidt trace of $L C L^*$ proves (11.3). $\square$

Repeated identical ranges saturate (11.2)–(11.3) exactly; overlap by itself is not a multiplicity obstruction. Polar minimization, however, is not a causal operation. If $S_1 = S_2 = 1/\sqrt{2}$ and $h = (0, f(\xi_1))$, the polar-minimal representative is

$$(f(\xi_1)/2, f(\xi_1)/2), \quad (11.4)$$

whose first coordinate is not $\mathcal{F}_0$ -measurable. Terminal polar compression therefore cannot promote a regular graph proof to a progressive one. This fires NG-2026-07-25-A13-TERMINAL-POLAR-CAUSAL-PROMOTION.

A revisit loop makes the termwise danger explicit. For $S_1 = S_2 = 1$ and $h = (t, -t)$, the terminal shift and terminal CM norm vanish but the time cost is $2t^2$. Along $W(a) = a^4/2$, the return increment is $-t^4/2$, whose ratio to the total time cost is $t^2/4$. The complete signed packet cancels the outgoing and return increments; estimating the return alone cannot be uniform.

There is nevertheless an exact causal entropy-union identity which avoids the polar representative. Let $\gamma = \bigotimes_k \gamma_k$ be standard Gaussian source measure and let

$$T_h(\xi) = \xi + h(\xi), \quad h_k = h_k(\xi_1, \dots, \xi_{k-1})$$

be bounded, smooth, and strictly triangular. The map T_h is a recursive diffeomorphism and its block-lower-triangular Jacobian has determinant one. For $\rho_h = (T_h)_{\#} \gamma$ this gives

$$H(\rho_h \mid \gamma) = \frac{1}{2} \mathbb{E}_{\gamma} \sum_k \|h_k\|^2. \quad (11.5)$$

Set $\mu_C = (S_{\pi})_{\#} \gamma$ and $\nu_h = (S_{\pi})_{\#} \rho_h$. Relative-entropy disintegration along the fibres of S_{π} yields

$$\frac{1}{2} \mathbb{E} \sum_k \|h_k\|^2 = H(\nu_h \mid \mu_C) + \mathcal{F}_h, \quad \mathcal{F}_h \geq 0, \quad (11.6)$$

where

$$\mathcal{F}_h = \int H(\rho_h(\cdot \mid S_{\pi} = z) \mid \gamma(\cdot \mid S_{\pi} = z)) \nu_h(dz).$$

Singular C causes no difficulty when the physical measures are read on $\text{Ran } S_{\pi}$. For the kernel loop $S_1 = S_2 = 1$ and $h = (t, -t)$, $\nu_h = \mu_C$ and $\mathcal{F}_h = t^2$: the entire source cost is retained as positive fibre entropy.

At $q = 10/9$, (11.6) and the entropy variational inequality give the exact diagnostic

$$\begin{aligned} & \mathbb{E} G_J(S_{\pi}(\xi + h)) + \frac{9}{20} \mathbb{E} \sum_k \|h_k\|^2 \\ & \geq -\frac{9}{10} \log \mathbb{E}_{\mu_C} e^{-(10/9) G_J} + \frac{9}{10} \mathcal{F}_h. \end{aligned} \quad (11.7)$$

This closes the overlap, kernel, and causal entropy accounting. It does not prove Nelson: the first term on the right is precisely the open Nelson free energy. Nor need a fixed finite strict-triangular class attain the full Gibbs variational infimum; with one source block it contains only translations.

The exact source-coordinate successor is the triangular covariance-union theorem. For every decomposition $C = \sum_k S_k S_k^*$ and every bounded smooth triangular $h_k = h_k(\xi_1, \dots, \xi_{k-1})$, prove uniformly that

$$\boxed{\mathbb{E} G_J \left(\sum_k S_k(\xi_k + h_k) \right) + \frac{9}{20} \mathbb{E} \sum_k \|h_k\|^2 \geq -C_A.} \quad (11.8)$$

R-081 temporal factorization shows that (11.8) is the class-faithful source coordinate form of H_A . It is invariant under coincident ranges and temporal refinement, preserves the cost of $\ker S_\pi$ loops, and does not require noncausal polar minimization. It remains unproved.

12 Negative flow and CAT(0) shortcuts do not close the frontier

Let γ_n be standard Gaussian measure and $A = \delta_\gamma b = x \cdot b - \operatorname{div} b = 2V$. First localize to a C^2 compactly supported b_R and solve

$$\dot{\Psi}_s = -b_R(\Psi_s), \quad \Psi_0 = I. \quad (12.1)$$

The flow is a global orientation-preserving diffeomorphism. With

$$D_t^R = \int_0^t s L_{b_R} A_R(\Psi_s X) ds, \quad A_R = \delta_\gamma b_R, \quad (12.2)$$

change of variables gives the exact identity

$$\boxed{\int e^{-tA_R - q\epsilon W} d\gamma = \mathbb{E} \exp\{D_t^R - q\epsilon W(\Psi_t X)\}, \quad q = 2t.} \quad (12.3)$$

Thus $q = 10/9$ corresponds to $t = 5/9$. The one-shot determinant zero has disappeared, but

$$\boxed{L_b(\delta_\gamma b) = |b|^2 + \delta_\gamma(Db b) + \operatorname{Tr}(Db)^2.} \quad (12.4)$$

Writing $Db = S + K$ gives $\operatorname{Tr}(Db)^2 = \|S\|_{\text{HS}}^2 - \|K\|_{\text{HS}}^2$; the signed curl obstruction remains.

For the untruncated negative production flow, radial dissipation gives forward completeness but not surjectivity. The exact scalar model $b(x) = x^3$ has

$$\Psi_t(x) = \frac{x}{\sqrt{1 + 2tx^2}}, \quad \operatorname{Im} \Psi_t = \left(-\frac{1}{\sqrt{2t}}, \frac{1}{\sqrt{2t}} \right). \quad (12.5)$$

Its Jacobian is positive everywhere, yet change of variables misses a positive high-field complement. Compact localization restores surjectivity but introduces the uncontrolled boundary term $-(\nabla \chi_R) \cdot b$. An exact Gibbs-tilt flow would require a correction w_s solving a weighted divergence equation under the still-open tilted law. The flow route is therefore not a noncircular Nelson proof.

Nor does conditional CAT(0) barycentering suffice. Scale the R-069 flat reset target by $\Phi_\lambda(z) = \lambda \sum_k z_k$. The fresh target remains exactly conditionally barycentered, while the paid ledger is

$$\left(-\frac{\lambda^2}{2} + \eta + 120\zeta \right) (J - 1) + 15\zeta. \quad (12.6)$$

For every fixed allocations η, ζ , choose $\lambda^2 > 2(\eta + 120\zeta)$; the ledger diverges negatively. Production spatial/source coupling is indispensable. These two failures are registered together as NG-2026-07-25-A13-NEGATIVE-FLOW-CAT0-SHORTCUT.

13 Attempt, failure, and evidence map

1. **Cumulative translated Z^6 majorant — already rejected.** It destroys saturation and repeats a critical budget.
2. **R-090 conservative coefficient compression — rejected and repaired.** The actual current uses J^T while ∇c uses J ; the exact two-field trace (4.4) retains both and preserves the regular closure.
3. **Normalized production lift — proved.** The Jacobian is a finite bounded polynomial and every secant Jacobian term contains one δn .
4. **Scalar superexponential tail as a vector theorem — rejected.** The anisotropic ratio (7.1) approaches one.
5. **Whole-product fractional largest input — proved.** It keeps coherent collisions and gives the $2^{-7j/30}$ root surplus.
6. **Regular physical-prefix one-use — proved.** The R-075 square function gives uniform prefix H^2/L^6 bounds and closes regular H_C .
7. **Terminal-prefix domination for revisits — rejected.** A return loop has zero terminal displacement but a large intermediate field.
8. **Separate r_C , J_D , covariance, and Schur estimates — retired.** They remain in one completed identity, with r_C in the perspective filtration and J_D in the full-frame baseline row.
9. **Terminal positive energy set aside — rejected.** Its Doob increments convert the signed Schur density into the augmented density (9.9), which vanishes exactly in the frozen case.
10. **Coefficient-conditioned moment-matched one reveal — proved.** Equation (9.11) is non-negative; general feedback reduces to a weighted conditional-covariance deficit and coefficient-reveal terms.
11. **Fixed or freshly adaptive derivative payment positivity — rejected.** The exact all-residual expectation is $-623/5440$ and varying R_k introduces (10.3).
12. **Covariance overlap as multiplicity loss — rejected.** Equations (11.2)–(11.3) are exact and overlap-uniform.
13. **Terminal polar compression as a causal proof — rejected.** The minimum representative (11.4) anticipates the first source coordinate.
14. **Triangular entropy union — proved as exact accounting.** Equation (11.6) stores overlap and kernel-loop cost in nonnegative fibre entropy, but (11.7) retains the open Nelson free energy.
15. **Negative flow — exact reduction, failed closure.** The determinant zero becomes a material derivative plus a high-field complement.
16. **CAT(0) barycenter alone — rejected.** The scaled reset model defeats every fixed allocation.

The remaining direct route is now

$$\boxed{H_C^{\text{reg}} \text{ closed} \quad + \quad H_N \text{ as (10.10)} \quad + \quad H_A \text{ as (11.8)} \quad \implies \quad \text{full packet/OVERLAP.}} \quad (13.1)$$

R-087 CORE then identifies a uniform full-OVERLAP bound with the $q = 10/9$ Nelson estimate. The two remaining analytic gates are coupled and may ultimately be proved by one triangular signed-packet theorem.

14 Devil's-advocate review

1. **Objection:** The normalized formula is only scalar. **DISMISSED.** Equations (2.3)–(2.9) hold for every symmetric six-real production generator with operator norm at most one.
2. **Objection:** Coherent Fourier collisions invalidate (5.6). **DISMISSED.** The finite products are formed before the fractional norm is taken; no product-family orthogonality is assumed.
3. **Objection:** The production Jacobian may be transposed in the chain rule. **UPHELD; PREDECESSOR CORRECTED.** Equations (3.5)–(3.8) separate actual b from $g = \nabla c$, and (4.4) uses the exact two-tail trace. No conservative identity is invoked.
4. **Objection:** The secant derivative contains an omitted $|a||Da|$ term. **DISMISSED.** The exact chain rule (3.9) has only $Dn(x+a)Da$ and the Dx -weighted derivative difference.
5. **Objection:** The exponent endpoint is critical. **DISMISSED.** The worst control total is $29/30$ and the worst root surplus is $2^{-7j/30}$.
6. **Objection:** Equation (1.1) covers arbitrary progressive revisits. **UPHELD as an overclaim.** The proof uses the R-075/R-079 physical one-shot prefix. General progression remains in (11.8).
7. **Objection:** The terminal $G^T \Theta_R(B_1)G$ energy may be ignored after isolating $\mathcal{K}_k$. **UPHELD as a sign error.** Its Doob increments are load-bearing in (9.9); the frozen augmented density is zero while the unaugmented density is negative.
8. **Objection:** The negative fixture disproves production H_N . **DISMISSED.** It lacks the complete spatial low/paid classification; it refutes only algebraic termwise positivity.
9. **Objection:** Covariance and entropy union prove H_A . **UPHELD as an overclaim.** They close overlap and fibre accounting, but (11.7) contains the open Nelson free energy and does not prove (11.8).
10. **Objection:** A positive flow Jacobian resolves Ramer. **DISMISSED.** Equation (12.5) has a positive Jacobian and still misses a positive-measure complement.

15 Executable verification and no-overclaim statement

The primary executable checks the normalized Jacobian and commutator, the transpose audit, corrected two-tail constants, secant/resolvent identities, both geometric denominators, fractional exponent inequalities, root and Young margins, physical-prefix finite models, matrix-perspective and augmented-density identities, the exact all-residual and covariance- deficit fixtures, covariance/entropy union, noncausal polar compression, negative-flow Liouville identity, and scaled CAT(0) ledger.

The non-importing independent executable uses random symmetric contractions, finite differences, independent matrix conditional ensembles, direct enumeration of the all-residual fixture, random overlapping source maps, flow quadrature, and a separate exponent reconstruction. The integrated verifier pins both scripts, the proof note, PDF contract, authority hashes, registry entries, claim surface, and all downstream false flags. Run

```
python codes/foundations/a13_classii_normalized_cartan_perspective_covariance_frontier_verify.py
```

R-092 closes only the regular one-shot Cartan H_C branch through the corrected two-field trace. It reduces H_N to the augmented perspective density and weighted conditional-covariance deficit, and gives exact causal entropy-union accounting for H_A . Complete signed NEAR, progressive/revisit assembly, full OVERLAP, Nelson, an interacting measure, and Sector A remain open. Tier stays T4.

Result footer

Result ID: A13-CLASSII-NORMALIZED-CARTAN-COMPENSATED-
PERSPECTIVE-TRIANGULAR-COVARIANCE-FRONTIER

Ledger: R-092.

Precise statement: The normalized production Jacobian is a bounded finite forest. The exact actual-b / gradient-c two-field trace and exhaustive fractional ledger close regular one-shot H_C with gap $2^{-(C-5)/2}$. The augmented perspective density closes frozen and coefficient-conditioned moment-matched one-reveal branches. Covariance union and strict-triangular entropy union are exact.

Scope: Finite cutoff and positive floor under pinned A1, R-075, R-079, R-084, R-087, R-089 conventions. H_C uses only the R-079 mutually orthogonal strict-past no-revisit one-shot class. Perspective uses a finite filtration. Entropy uses bounded smooth finite-dimensional strict-triangular shifts.

Dependencies: A1; R-063; R-075; R-079; R-084; R-087; R-089; audited R-090 and R-091 boundary identities.

Evidence grade: ANALYTIC, EXACT, EXECUTED, BOUNDARY.

Reproduction command: python codes/foundations/a13_classii_normalized_cartan_perspective_covariance_frontier_verify.py

Expected output: Primary 83/83; independent 46/46; integrated 147/147; aggregate 276/276; exit zero.

Negative results: AUDIT-2026-07-25-A13-R090-CONSERVATIVE-TRANSPOSE;
NG-2026-07-25-A13-SCALAR-SUPEREXPONENTIAL-VECTOR-UNIFORMITY;
NG-2026-07-25-A13-PERSPECTIVE-INNOVATION-TERMWISE-POSITIVITY;
NG-2026-07-25-A13-TERMINAL-POLAR-CAUSAL-PROMOTION;
NG-2026-07-25-A13-NEGATIVE-FLOW-CATO-SHORTCUT.

Falsification gate: Failure of the transpose audit, two-field trace, constants 16 or 32, any of the seven control monomials, 29/30 worst exponent, 1/30 Young slack, augmented perspective identities, exact fixtures, covariance/entropy union, hashes, PDF, or counts.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No general progressive/revisit H_C, complete H_N, progressive H_A, controlled-shell one-use, complete REG, full OVERLAP, Nelson, floor removal, interacting measure, tier promotion, or Sector-A closure.

Next required action: Prove the augmented weighted conditional-covariance and multistep derivative-feedback H_N bound; prove triangular progressive/revisit H_A; then assemble full OVERLAP and invoke R-087 CORE for Nelson.